OTTO

TURKISH COFFEE MACHINE

A solution that combines the fast pace of the office with the traditional Turkish coffee ritual, balancing speed and experience

Milestone 1 : Empathize

AI ROLE: Clarifier

USER INSIGHT

User research shows that Turkish coffee is not just a beverage, but also a social catalyst. While current machines focus on speed, they eliminate the psychological satisfaction of the "cezve" form, the social chats during preparation, and the relaxing effect of the ritual. Users seek to socialize and connect with the process during their breaks.

KEYWORDS

- *Social Catalyst*
- *Psychological Satisfaction*
- *Cezve Archetype*
- *Sensory Connection*

PROMPT

*"How does the **traditional form** of a **'cezve'** affect the user's perception of taste and experience? Analyze the role of Turkish coffee preparation as **a social ritual** in an office break culture."*

AI RESPONSE

*AI noted that the cezve form strengthens the perception of **"authenticity,"** and unlike mechanical machines, the waiting time during preparation creates a valuable **"social window"** for employees.*

DESIGN DECISION

AI noted that the cezve form strengthens the perception of **"authenticity,"** and unlike mechanical machines, the waiting time during preparation creates a valuable **"social window"** for employees.

TRANSFORMATION

User Problem (Social disconnection and soulless automation) → Design Opportunity (Social Interaction and Ritual Station)

Milestone 2 : Define

AI ROLE: Clarifier

PROBLEM STATEMENT

In the traditional preparation process, manually measuring coffee, sugar, and water for each serving leads to inefficient use of limited break time. On the other hand, while fully automatic systems shorten this time, they eliminate the user's influence over "foam management" and "portioning precision"—the most critical quality indicators of Turkish coffee. The problem is to accelerate time-consuming logistical steps (dosing) through device precision while keeping the user involved only during the high-value ritual moments.

KEYWORDS

- *Precision Dosing*
- *Time Efficiency*
- *Foam Quality Optimization*

PROMPT

"How can we automate the repeated measurement of ingredients (water, coffee, sugar) to save time, and how can we use automatic portioning to guarantee the perfect foam height that the user will enjoy? And while doing all this, we shouldn't disconnect the user from the rituals of Turkish coffee."

AI RESPONSE

AI proposed the concept of "Logistic Automation": Storage and dosing should be performed with high precision by the machine to maintain the foam standard, while the user should focus only on the "cooking interaction" and "mixing process," thus spending their break time efficiently and productively

DESIGN DECISION

Decision: The burden of adding ingredients each time has been eliminated; an ideal foam base is prepared through the device's portioning precision, enabling the user to focus only on the ritual.

TRANSFORMATION

Problem (Inefficient preparation and inconsistent foam quality) → Design Statement (Precision-Dosed Ritual Station)

Milestone 3 : Ideate

AI ROLE: Re-framer

USER INSIGHT

Users want to avoid the hassle of measuring ingredients every time but are afraid of losing complete control to the machine. Seeing the amount of coffee, witnessing the foam formation, and intervening through stirring are indispensable needs for psychological satisfaction.

KEYWORDS

- *Ritualistic Efficiency*
- *Archetype Modernization*
- *Seamless Integration*

PROMPT

"Design a form that bridges the gap between high-speed office needs and slow-paced coffee rituals. Propose a system where the boring logistics are hidden, but the iconic 'cezve' remains the hero of the stage."

AI RESPONSE

AI suggested an architecture that dispenses ingredients (water, coffee, sugar) directly into the pot from a "Bridge" mechanism above. This ensures that logistical steps are completed rapidly in the background, while the pot and brewing process remain the visual and physical focus of the user.

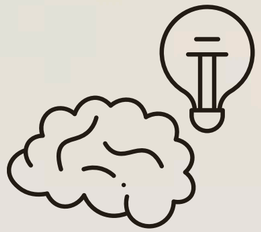
DESIGN DECISION

Accepted: Transparent modular tubes integrated into the machine. This allows for long-term use with a single refill while enabling visual tracking of remaining ingredients.

- **Accepted:** Transparent borosilicate glass pot. Chosen to ensure the visibility of the brewing process and maintain connection with the ritual.
- **Accepted:** User Stirring Experience. The stirring process was deliberately assigned to the user to preserve the ritual in a modern environment.
- **Rejected:** Closed, opaque containers and fully automatic (non-stirring) systems where ingredients and the brewing process are hidden.

TRANSFORMATION

Idea (Office Speed vs. Tradition) → Solution (Ritual Staged with a Modern Dosing Bridge)



Milestone 4 : Prototype

AI ROLE: Expander

USER INSIGHT

"Hygiene is critical in an office environment. The device parts must be easy to clean and deliver the same taste every time. Also, the containers must be secure to keep the ingredients fresh."

KEYWORDS

- *Detachable Containers*
- *Auger-Dosing System*
- *IR Foam Monitoring*

PROMPT

"Propose a mechanical system that uses detachable containers for easy cleaning. How can we integrate an auger system to dose ingredients precisely from these removable units?"

AI RESPONSE

AI suggested a detachable locking mechanism for each container and an Auger (Screw) system integrated into the base. This structure allows containers to be easily removed for refilling or cleaning; when replaced, they automatically sync with the motor for precise dosing.

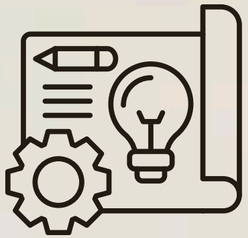
DESIGN DECISION

Accepted: Detachable transparent tubes for practical cleaning and refilling.

- **Accepted:** Auger motor system that provides precise portioning and works in full sync with the containers.
- **Accepted:** IR (Infrared) Sensor hidden under the bridge to prevent overflow.
- **Rejected:** Fixed containers that make cleaning difficult and open-mouth systems with hygiene risks.

TRANSFORMATION

Technical Problem (Difficult cleaning and staleness risk) → Technical Solution (Detachable Hygienic Containers and Precision Dosing)



Milestone 5 : Test

AI ROLE: Critique

USER INSIGHT

"In our boutique office of 4-5 people, everything is very intimate. We want the coffee machine to be a decorative piece that harmonizes with our desks or lounge area, not a kitchen robot. It's also important for us that it creates a space where we can chat while preparing coffee."

KEYWORDS

- *Boutique Office Harmony*
- *Social Hub for Team*
- *Warm Minimalism*

PROMPT

"Evaluate the final design for a small office of 4-5 people. Does the white-copper-walnut palette reflect a premium lifestyle object? How does this color transition affect the user's perception of the ritual?"

AI RESPONSE

AI noted that the purity of white and the traditional warmth of copper create a perfect contrast. It emphasized that the walnut handle detail transforms a digital/technological device into a "domestic and intimate" object, which meets the need for a "peaceful break" in boutique offices.

DESIGN DECISION

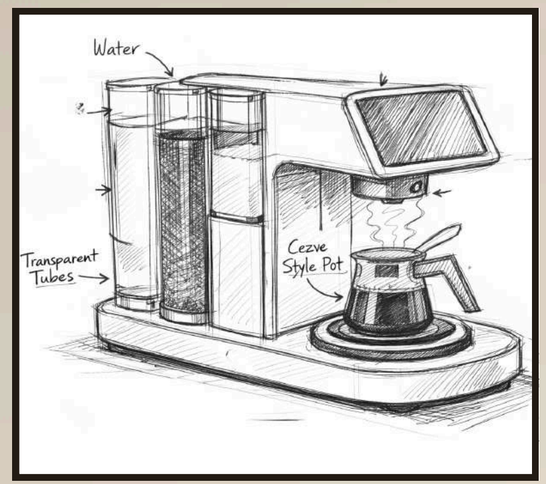
Accepted: Walnut Wood and Satin White textures suitable for the warmth of small teams.

- **Accepted:** An open and accessible design language that turns the machine into a "social island."
- **Accepted:** A compact structure that meets the daily consumption of 4-5 people while taking up minimal desk space.
- **Rejected:** Cold and massive plastic bodies designed for large corporate offices.

TRANSFORMATION

Result (Industrial Lab Look) → Design (Warm and Luxury Ritual Station for Boutique Offices)

VISUAL PROCESS



DESIGN DECISIONS

FINAL DESIGN CONCEPT

PERSPECTIVE VIEW

Digital Touchscreen

Glass Coffee Container

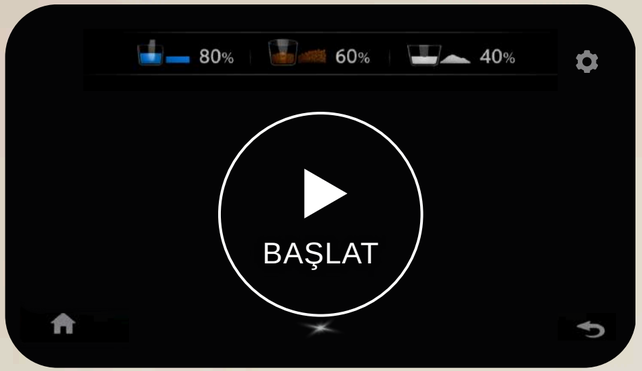
- *maximum 4 cups of coffee*

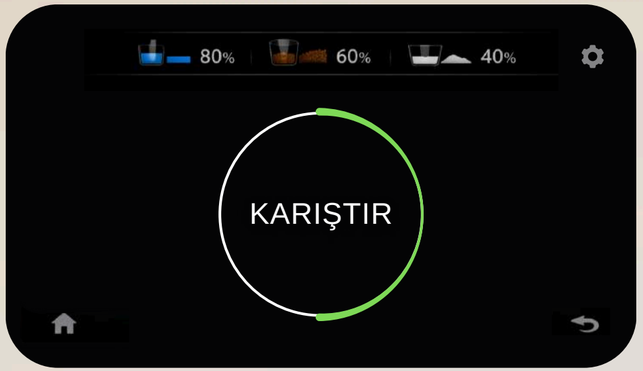
Vacuum Tubes

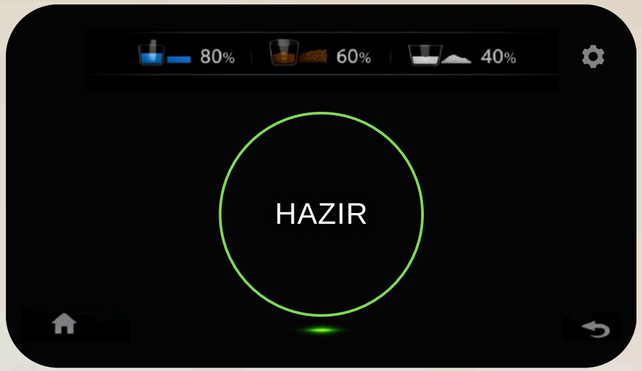
BACK VIEW

DESIGN DECISIONS

1







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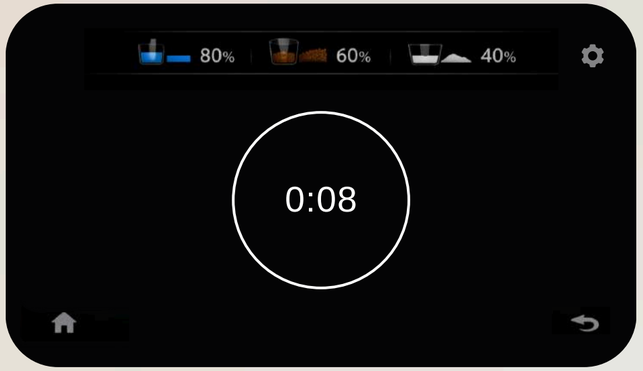

1) The Turkish coffee brewing process

2) Self-Cleaning Process

USER SCENARIO

- **Selection & Auto-Prep The user plugs in Otto, and the device activates. On the touch screen, the user selects "Turkish Coffee," then chooses the portion size and sugar level before hitting start. Otto automatically dispenses the precise amount of coffee, sugar, and water into the pot.**

- **Interactive Stirring Shortly after the brewing begins, the device emits the first "beep-beep" alert, and the screen displays "Stir." The user lifts the pot and stirs the coffee, actively participating in the ritual. Once the pot is placed back and the user presses the "Continue" button, the brewing process resumes.**

- **Smart Brewing & Completion The user enjoys watching the coffee foam and rise through the transparent glass pot. While Otto's sensors optimize the foam level, a second "beep-beep" alert sounds to notify the user that the coffee is perfectly brewed and ready.**

- **Serving Gripping the ergonomic walnut handle, the user serves the perfectly consistent coffee. Otto handles the logistical precision while ensuring the user remains connected to the soul of the ritual.**

MY REFLECTIONS

Throughout this project, I utilized AI as a "studio partner" to filter my ideas and optimize technical solutions. My primary motivation in designing Otto was to resolve the conflict between the need for speed in boutique offices and the ancient rituals of Turkish coffee. Through my dialogue with AI, I discovered how technology, rather than destroying the ritual, can enhance its quality by taking over "logistical burdens" like dosing and freshness management. This process taught me that design is not just about providing a solution, but about preserving cultural habits by adapting them to the modern world.